In [1]:
```python
import pandas as pd
import pandas as np
```

In [2]:
```python
df = pd.read_csv("uncleaned_cafe_sales.csv")
```

In [3]:
```python
df.head()
```

Out[3]:

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
0	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
1	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
2	TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
3	TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
4	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11

In [4]:
```python
df.shape
```

Out[4]:
```
(10000, 8)
```

In [5]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 8 columns):
 #   Column            Non-Null Count  Dtype 
---  ------            --------------  ----- 
 0   Transaction ID    10000 non-null  object
 1   Item              9667 non-null   object
 2   Quantity          9862 non-null   object
 3   Price Per Unit    9821 non-null   object
 4   Total Spent       9827 non-null   object
 5   Payment Method    7421 non-null   object
 6   Location          6735 non-null   object
 7   Transaction Date  9841 non-null   object
dtypes: object(8)
memory usage: 625.1+ KB
```

In [6]:
```python
df.isna().sum()
```

Out[6]:
```
Transaction ID         0
Item                 333
Quantity             138
Price Per Unit       179
Total Spent          173
Payment Method      2579
Location            3265
Transaction Date     159
dtype: int64
```

In [7]:
```python
df.describe(include="object")
```

Out[7]:

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
count	10000	9667	9862	9821	9827	7421	6735	9841
unique	10000	10	7	8	19	5	4	367
top	TXN_9226047	Juice	5	3.0	6.0	Digital Wallet	Takeaway	UNKNOWN
freq	1	1171	2013	2429	979	2291	3022	159

In [8]:
```
df = df.dropna()
```

In [9]:
```
df
```

Out[9]:

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Trans
0	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023
1	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023
2	TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023
3	TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023
4	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023
...	...	...	...	...	...	...	...	
9984	TXN_3142496	Smoothie	UNKNOWN	4.0	4.0	Cash	Takeaway	2023
9986	TXN_2858441	Sandwich	2	4.0	8.0	Credit Card	In-store	2023
9991	TXN_3897619	Sandwich	3	4.0	12.0	Cash	Takeaway	2023
9992	TXN_2739140	Smoothie	4	4.0	16.0	UNKNOWN	In-store	2023
9999	TXN_6170729	Sandwich	3	4.0	12.0	Cash	In-store	2023

4550 rows × 8 columns

In [10]:
```
df.duplicated().sum()
```

Out[10]:
```
np.int64(0)
```

In [11]:
```
df = df.drop_duplicates()
```

In [12]:
```
df
```

Out[12]:

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Trans
0	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023
1	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023
2	TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023
3	TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023
4	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023
...	...	...	...	...	...	...	...	
9984	TXN_3142496	Smoothie	UNKNOWN	4.0	4.0	Cash	Takeaway	2023
9986	TXN_2858441	Sandwich	2	4.0	8.0	Credit Card	In-store	2023
9991	TXN_3897619	Sandwich	3	4.0	12.0	Cash	Takeaway	2023
9992	TXN_2739140	Smoothie	4	4.0	16.0	UNKNOWN	In-store	2023
9999	TXN_6170729	Sandwich	3	4.0	12.0	Cash	In-store	2023

4550 rows × 8 columns

In [13]:
```
pd.Series(df["Transaction
ID"].unique().tolist()).str.startswith("TXN").value_counts()
```

Out[13]:
```
True    4550
Name: count, dtype: int64
```

In [14]:
```
df["Quantity"].unique()
```

Out[14]:
```
array(['2', '4', '3', '5', 'ERROR', '1', 'UNKNOWN'], dtype=object)
```

In [15]:
```
df.columns
```

Out[15]:
```
Index(['Transaction ID', 'Item', 'Quantity', 'Price Per Unit', 'Total Spent',
       'Payment Method', 'Location', 'Transaction Date'],
      dtype='object')
```

In [16]:
```
df.isin(["UNKNOWN","ERROR"]).sum()
```

Out[16]:
```
Transaction ID        0
Item                295
Quantity            146
Price Per Unit      168
Total Spent         155
Payment Method      346
Location            468
Transaction Date    128
dtype: int64
```

In [17]:
```
df = df[~df.isin(["UNKNOWN","ERROR"]).any(axis=1)]
```

In [18]:
```
df
```

Out[18]:

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
0	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
1	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
4	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
10	TXN_2548360	Salad	5	5.0	25.0	Cash	Takeaway	2023-11-07
12	TXN_7619095	Sandwich	2	4.0	8.0	Cash	In-store	2023-05-03
...	...	...	...	...	...	...	...	...
9975	TXN_9668108	Cake	1	3.0	3.0	Cash	In-store	2023-01-20
9979	TXN_9933628	Smoothie	5	4.0	20.0	Cash	In-store	2023-07-20
9986	TXN_2858441	Sandwich	2	4.0	8.0	Credit Card	In-store	2023-12-14
9991	TXN_3897619	Sandwich	3	4.0	12.0	Cash	Takeaway	2023-02-24
9999	TXN_6170729	Sandwich	3	4.0	12.0	Cash	In-store	2023-11-07

3089 rows × 8 columns

In [19]:
```
(df == "ERROR").sum()
```

Out[19]:
```
Transaction ID      0
Item                0
Quantity            0
Price Per Unit      0
Total Spent         0
Payment Method      0
Location            0
Transaction Date    0
dtype: int64
```

In [20]:
```
df.dtypes
```

Out[20]:
```
Transaction ID      object
Item                object
Quantity            object
Price Per Unit      object
Total Spent         object
Payment Method      object
Location            object
Transaction Date    object
dtype: object
```

In [21]:
```
df.dtypes
```

Out[21]:
```
Transaction ID      object
Item                object
Quantity            object
Price Per Unit      object
Total Spent         object
Payment Method      object
Location            object
Transaction Date    object
dtype: object
```

In [22]:
```
df["Quantity"] = pd.to_numeric(df["Quantity"],
errors="coerce").astype("Int64")
```

```
C:\Users\rgbro\AppData\Local\Temp\ipykernel_14860\2830958091.py:1: SettingWithCop
yWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stabl
e/user_guide/indexing.html#returning-a-view-versus-a-copy
  df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce").astype("Int64")
```

In [23]:
```
df.dtypes
```

Out[23]:
```
Transaction ID      object
Item                object
Quantity             Int64
Price Per Unit      object
Total Spent         object
Payment Method      object
Location            object
Transaction Date    object
dtype: object
```

In [24]:
```
import warnings
warnings.filterwarnings("ignore")
```

In [25]:
```
df["Price Per Unit"] = pd.to_numeric(df["Price Per Unit"], errors="coerce")
df["Total Spent"] = pd.to_numeric(df["Total Spent"], errors="coerce")
```

In [26]:
```
df.dtypes
```

Out[26]:
```
Transaction ID       object
Item                 object
Quantity              Int64
Price Per Unit      float64
Total Spent         float64
Payment Method       object
Location             object
Transaction Date     object
dtype: object
```

In [27]:
```
df["Transaction Date"] = pd.to_datetime(df["Transaction Date"],
errors="coerce")
```

In [28]:
```
df.dtypes
```

Out[28]:

```
Transaction ID              object
Item                        object
Quantity                     Int64
Price Per Unit             float64
Total Spent                float64
Payment Method              object
Location                    object
Transaction Date    datetime64[ns]
dtype: object
```

In [29]:

```
df
```

Out[29]:

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
0	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
1	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
4	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
10	TXN_2548360	Salad	5	5.0	25.0	Cash	Takeaway	2023-11-07
12	TXN_7619095	Sandwich	2	4.0	8.0	Cash	In-store	2023-05-03
...	...	...	...	...	...	...	...	...
9975	TXN_9668108	Cake	1	3.0	3.0	Cash	In-store	2023-01-20
9979	TXN_9933628	Smoothie	5	4.0	20.0	Cash	In-store	2023-07-20
9986	TXN_2858441	Sandwich	2	4.0	8.0	Credit Card	In-store	2023-12-14
9991	TXN_3897619	Sandwich	3	4.0	12.0	Cash	Takeaway	2023-02-24
9999	TXN_6170729	Sandwich	3	4.0	12.0	Cash	In-store	2023-11-07

3089 rows × 8 columns

In [30]:

```
df.sort_values("Transaction ID").reset_index(drop = True)
```

Out[30]:

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
0	TXN_1000555	Tea	1	1.5	1.5	Credit Card	In-store	2023-10-19
1	TXN_1002457	Cookie	5	1.0	5.0	Digital Wallet	Takeaway	2023-09-29
2	TXN_1004184	Smoothie	1	4.0	4.0	Credit Card	In-store	2023-05-18
3	TXN_1004563	Tea	5	1.5	7.5	Credit Card	In-store	2023-10-28
4	TXN_1005331	Coffee	1	2.0	2.0	Digital Wallet	Takeaway	2023-11-04
...	...	...	...	...	...	...	...	...
3084	TXN_9989719	Smoothie	2	4.0	8.0	Credit Card	Takeaway	2023-11-26
3085	TXN_9991225	Cake	5	3.0	15.0	Credit Card	Takeaway	2023-02-03
3086	TXN_9992148	Tea	3	1.5	4.5	Cash	Takeaway	2023-01-14
3087	TXN_9996195	Juice	5	3.0	15.0	Digital Wallet	In-store	2023-03-16
3088	TXN_9999113	Juice	4	3.0	12.0	Cash	Takeaway	2023-05-27

3089 rows × 8 columns

In [32]:
```
df = df.sort_values("Transaction ID").reset_index(drop = True)
```

In [33]:
```
df
```

Out[33]:

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
0	TXN_1000555	Tea	1	1.5	1.5	Credit Card	In-store	2023-10-19
1	TXN_1002457	Cookie	5	1.0	5.0	Digital Wallet	Takeaway	2023-09-29
2	TXN_1004184	Smoothie	1	4.0	4.0	Credit Card	In-store	2023-05-18
3	TXN_1004563	Tea	5	1.5	7.5	Credit Card	In-store	2023-10-28
4	TXN_1005331	Coffee	1	2.0	2.0	Digital Wallet	Takeaway	2023-11-04
...	...	...	...	...	...	...	...	...
3084	TXN_9989719	Smoothie	2	4.0	8.0	Credit Card	Takeaway	2023-11-26
3085	TXN_9991225	Cake	5	3.0	15.0	Credit Card	Takeaway	2023-02-03
3086	TXN_9992148	Tea	3	1.5	4.5	Cash	Takeaway	2023-01-14
3087	TXN_9996195	Juice	5	3.0	15.0	Digital Wallet	In-store	2023-03-16
3088	TXN_9999113	Juice	4	3.0	12.0	Cash	Takeaway	2023-05-27

3089 rows × 8 columns

In [35]:
```
df.to_excel("Cafe_Sales_Cleaned.xlsx", index=False)
```

In [ ]: